

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

Department of Electrical Engineering - ICDT

PROJECT REPORT

UGV – Part 1: Hardware Assembly and Integration

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1 Introduction

This project focuses on the development of a mobile phone based voice controlled Unmanned Ground Vehicle (UGV). Part-1 primarily deals with the mechanical assembly and hardware integration of the vehicle. It involves mounting the DC motors, assembling the acrylic chassis, and integrating the ESP32, L293D motor driver and power supply. The objective is to establish a functional hardware platform for subsequent implementation of voice-controlled operation.

2 Project Objectives

- To inspect and utilise the pre-assembled acrylic chassis with DC geared motors and wheels as the mechanical foundation of the UGV.
- To integrate the ESP32 development board and L293D motor driver onto the perfboard.
- To establish the necessary electrical connections using jumper wires.
- To incorporate the two 4 V batteries and battery holder as the power supply for the UGV.
- To achieve a functional hardware assembly and verify the basic operation of the DC motors.
- To ensure secure mounting and proper electrical connections of all hardware components.

3 Component Overview

The ugv is implemented using discrete hardware components distributed for the laboratory project. This section details the list of materials used in the physical assembly.

3.1 List of Materials

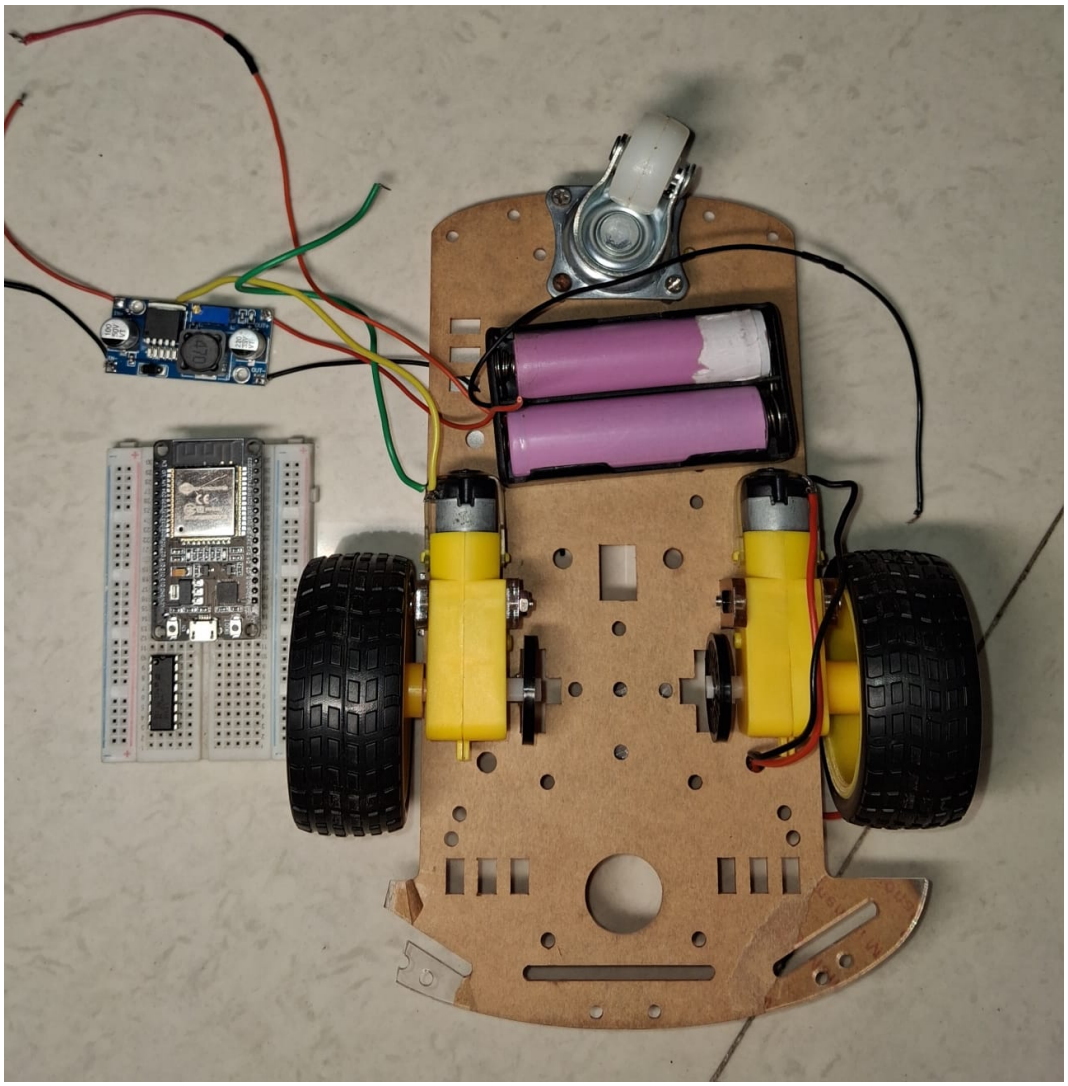


Figure 1: Components for ugv

Table 1 outlines the specific parts received and integrated into the bread board setup.

Table 1: List of Components

S.No.	Component Name	Quantity
1	Acrylic chassis	1
2	DC geared motors	2
3	Bread Board	1
4	ESP32 dev board	1
5	L293D motor driver IC	1
6	DC-DC buck converter	1
7	Batteries(3.7V , 2200mAH)	2
8	Capacitor(100 μF/25V/85°C)	1
9	Connecting Wires and Solder	as needed

3.2 Overview of Components

- **`Acrylic Chassis:`** It serves as the main structural frame of the UGV. It provides support for the motors, wheels, batteries and electronic components.
- **`ESP32 Development Board:`** The ESP32 is a microcontroller development board that acts as the main control unit of the UGV. It processes movement instructions and generates control signals through its GPIO(General Purpose Input/Output) pins. These signals are sent to the L293D motor driver to control the two DC geared motors. It also has built-in Wi-Fi and Bluetooth capabilities, making it suitable for wireless control applications.
- **`L293D Motor Driver IC:`** The L293D is a dual H-bridge motor driver IC used to control two DC motors independently. The ESP32 cannot directly provide the current required to drive the motors. Hence, the L293D acts as an interface between the ESP32 and the motors. By changing the direction of current through the motor terminals, it allows the motors to rotate in both directions. This enables the UGV to move forward, backward and turn left or right.

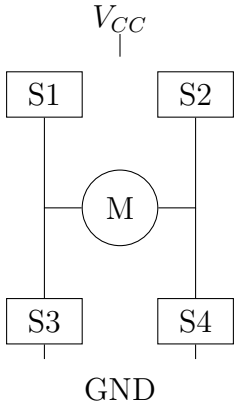


Figure 2: Basic H-bridge arrangement for controlling a DC motor

In the above diagram, S1, S2, S3 and S4 represent the four switching elements of an H-bridge circuit. By activating opposite pairs of switches, the direction of current through the motor can be reversed, allowing it to rotate in either direction. Here, M represents the DC motor connected to the H-bridge circuit.

- **`DC-DC Buck Converter:`** It is a step-down voltage regulator. It reduces the input DC voltage to a lower, regulated voltage suitable for powering electronic components.

4 Working Principle

- The UGV is powered by two rechargeable lithium-ion batteries, each rated at 3.7 V, connected in series. When connected in series, their voltages add up, providing a nominal output voltage of 7.4 V. This battery pack acts as the primary source of electrical energy for the entire system.

- The output from the battery pack is supplied to a DC-DC buck converter, which steps down the input voltage and is adjusted to provide a regulated output of 5 V. This conversion is necessary because the battery pack provides a nominal voltage of 7.4 V, whereas the ESP32 development board requires a lower input voltage. The 5 V output is supplied through the 5V/VIN input of the development board.

- A 100 μF capacitor is also connected in the power supply circuit to help maintain a stable supply voltage. During the initial testing, the ESP32 was observed to disconnect from the mobile phone after some time. After incorporating the capacitor, the disconnections became less frequent, indicating an improvement in the stability of the system.

- The regulated power supply is used to power the ESP32 development board. When a movement instruction is given, the ESP32 processes it and sends the corresponding control signals to the L293D motor driver. These signals determine the required movement of the vehicle.

- Based on the control signals received, the L293D motor driver supplies the required voltage and current to the two DC geared motors, controlling their direction of rotation. The gear mechanism reduces the rotational speed while increasing the available torque, making the motors suitable for driving the UGV.

- The two motors drive the wheels attached to the acrylic chassis. When both motors rotate in the same direction, the vehicle moves forward or backward. When the motors rotate in opposite directions or at different speeds, the vehicle can turn left or right. By controlling the operation of both motors independently, the UGV can perform the required movements.

In summary, we can say the working principle as:

7.4 V DC Battery Supply → DC-DC Buck Converter → Regulated 5 V DC Supply(with 100 μF Capacitor for Voltage Stabilisation) → ESP32 Controller → L293D Motor Driver IC → DC Geared Motors → UGV Movement

4.1 Circuit Diagram

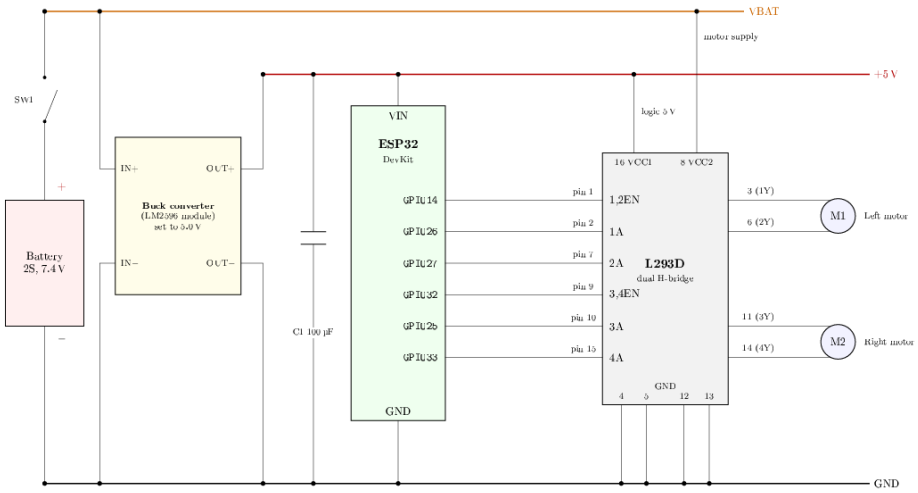
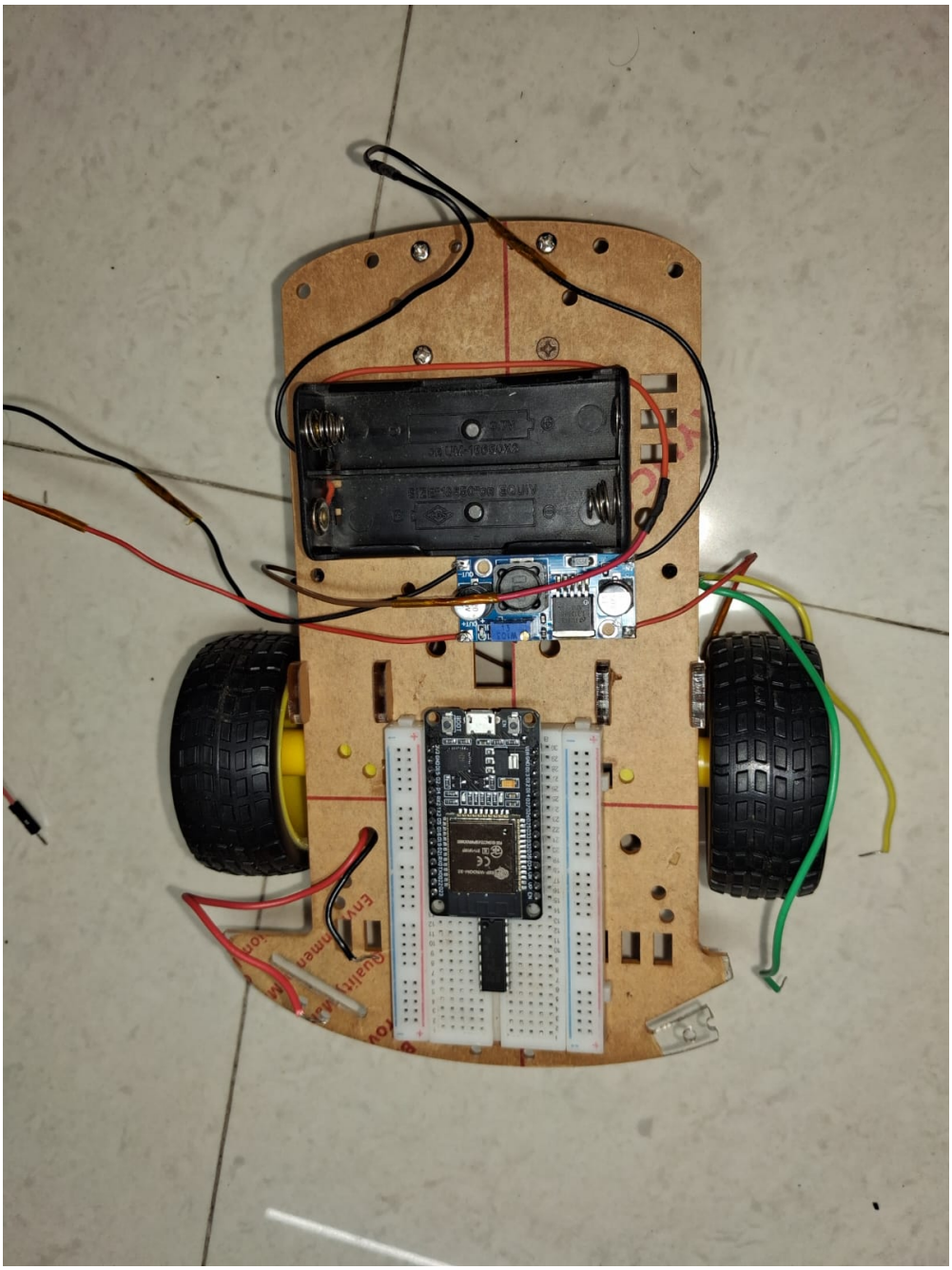


Figure 3: Circuit Diagram of UGV

5 Circuit Assembly on Bread Board

5.1 Component Placement

The components were placed on the bread board, according to the circuit diagram,keeping the connections as compact and organized as possible and the assembled frame looks like this



5.2 Electrical Connections and Integration

After deciding the component arrangement, the necessary electrical connections were established using jumper wires. The ESP32 development board, L293D motor driver IC, DC-DC buck converter and other components were connected according to the circuit arrangement. These connections formed the electrical system required for controlling the two DC geared motors.

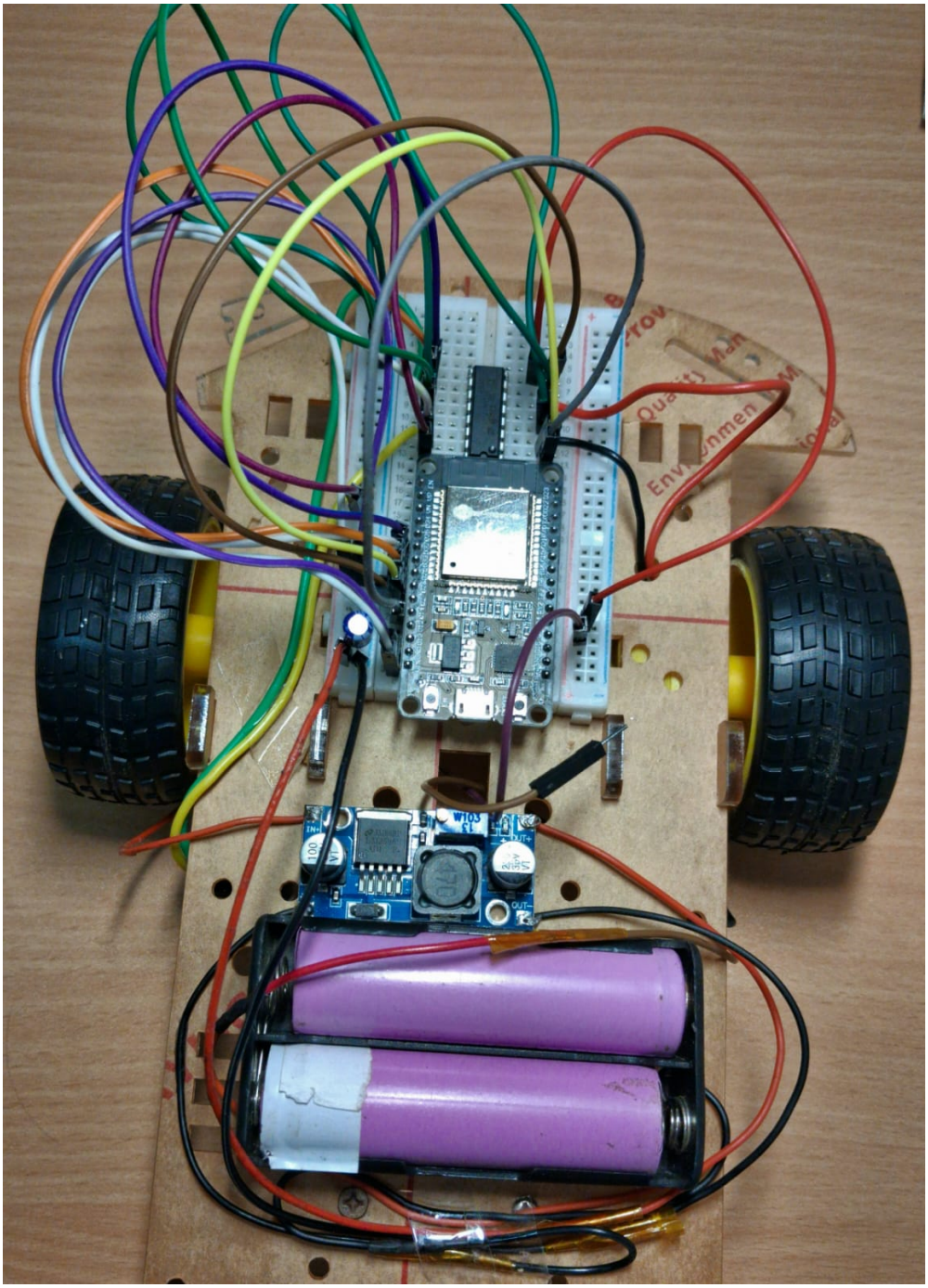


Figure 4: UGV hardware assembly after connecting jumper wires

6 Testing and Voltage Measurements

6.1 Battery and Power Supply Measurements

After completing the hardware assembly, basic voltage checks were carried out to examine the power supply of the UGV.

The two rechargeable lithium-ion batteries, each rated at 3.7 V and 2200 mAh, were measured individually. Each battery showed a voltage close to 4 V. When connected in series, the batteries provide a nominal output voltage of 7.4 V.

The DC-DC buck converter was adjusted to provide a regulated output of 5 V for powering the ESP32 development board. A 100 μF capacitor was also incorporated into the power supply circuit to help reduce voltage fluctuations and improve supply stability.

These readings helped us confirm that the circuit was working, safely, as expected.

6.2 Initial Testing and Observations

During the initial testing, the ESP32 was observed to disconnect from the mobile phone after some time. To address this issue, a 100 μF capacitor was added to the power supply circuit.

After incorporating the capacitor, the disconnections became less frequent. This suggested that fluctuations in the power supply might have contributed to the connectivity issue. The capacitor helped improve the stability of the system during operation.

7 Problems Faced

7.1 ESP32 Bluetooth Connectivity Issues

During the initial testing of the UGV, the DC motors were functioning, but the Bluetooth connection between the ESP32 and the mobile phone was observed to flicker intermittently. Initially, a voltage regulator IC was used to obtain the required supply voltage. Although the motors operated, the Bluetooth connection was unstable.

We suspected that the issue could be related to fluctuations in the power supply. A voltage regulator IC may experience voltage drops or heating depending on the input voltage and current demand. Since the motors draw varying amounts of current during operation, these variations could have affected the supply stability of the ESP32.

To address this issue, the initial voltage regulation arrangement was replaced with a DC-DC buck converter adjusted to provide a 5 V output. Additionally, a 100 μF capacitor was incorporated into the power supply circuit to help reduce short-term voltage fluctuations. Following these changes, the Bluetooth disconnections became less frequent, resulting in improved connectivity.

7.2 Faulty Motor Driver IC

During the initial assembly and testing, the motors did not respond as expected to the control signals. We suspected that the problem could be due to incorrect wiring or a faulty motor driver IC. The connections and power supply were checked to identify the source of the problem. After troubleshooting, the motor driver IC was identified as faulty and was replaced. This resolved the issue and allowed the motor control circuit to function properly.

7.3 Faulty DC Geared Motor

During testing, one of the DC geared motors was found to rotate unusually slowly and occasionally get stuck. This affected the movement of the UGV and indicated that the motor was not functioning properly. After identifying the issue, the faulty motor was exchanged for another one. The replacement motor operated more reliably and helped ensure proper movement of the vehicle.

7.4 Absence of an On/Off Switch

The chassis frame supplied to us did not have a provision for an integrated power switch. As a result, turning off the UGV required manually disconnecting the power connection. This made switching the vehicle on and off slightly inconvenient during testing and operation. Although the system could still be powered by connecting and removing the batteries, the absence of a dedicated switch remained a minor limitation of the hardware assembly.

8 Learnings and Experience

Through the hardware assembly and integration of the UGV, we gained practical experience in the following areas:

- **Microcontroller and Motor Driver Integration:** Understood how the ESP32 communicates with the L293D motor driver to control the DC geared motors.
- **Power Supply Management:** Learned how to connect lithium-ion batteries in series and use a DC-DC buck converter to obtain the required supply voltage.
- **Role of Capacitors:** Understood how a capacitor can help reduce short-term voltage fluctuations and improve power supply stability.
- **Hardware Troubleshooting:** Gained experience in identifying faulty components, replacing defective ICs and motors, and testing the circuit again.
- **Wireless Connectivity:** Learned the importance of a stable power supply for maintaining reliable Bluetooth communication with the ESP32.
- **Component Assembly:** Gained practical experience in mounting components on the chassis and establishing electrical connections using jumper wires.

9 Completion and Conclusion

The first phase of the UGV project marks the completion of its basic hardware assembly and integration. The assembled platform establishes the necessary foundation for implementing the control system in the subsequent phase.

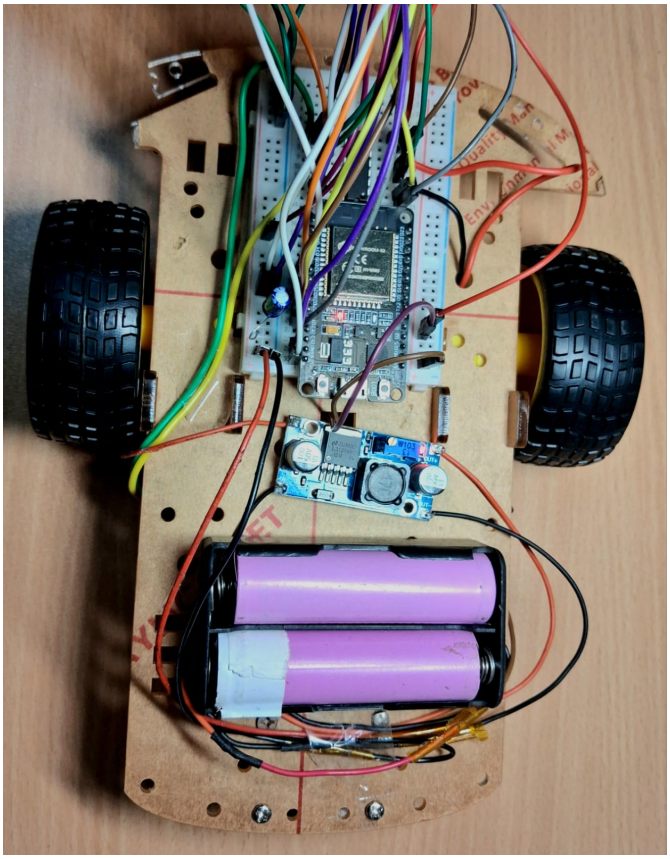


Figure 5: working UGV

Through this phase, we gained practical experience in hardware integration and troubleshooting, while addressing the challenges encountered during assembly. The project is now ready to proceed towards the implementation of voice-controlled operation.